

Nikita Starodubcev

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Google Scholar: [Nikita Starodubcev](#)

Languages: English (fluent), Russian (native).

RESEARCH

Efficient generative modeling for computer vision, with a focus on diffusion and few-step models for image generation and editing. Interests include model behavior analysis, inference acceleration, distillation, and practical visual generative systems.

EDUCATION

HSE University, Ph.D. Student in Computer Science Nov 2023 – Nov 2026 (expected)
Supervisor: [Artem Babenko](#), GPA: 5.0/5.0
Thesis: Efficient Diffusion Models for Computer Vision Tasks

ITMO University, M.S. in Computer Science Sep 2021 – Jul 2023
Supervisor: [Nikolay Nikitin](#), GPA: 5.0/5.0
Thesis: Generative Design of Physical Objects with a Modular Framework

Saint Petersburg State University, B.S. in Physics Sep 2017 – Jul 2021
Supervisor: [Evgeny Yarevsky](#), GPA: 4.8/5.0
Thesis: Machine Learning for Quantum-Mechanical Resonance Identification

EXPERIENCE

Adobe Research, *Research Intern* May 2025 – Aug 2025
Worked on improving the generation quality of diffusion models.
Hosted by [Zhe Lin](#), [Zongze Wu](#), and [Daniil Pakhomov](#)

Yandex Research, *Research Scientist* Nov 2022 – Present
Working on accelerating diffusion models for image generation and editing.

AIM Club ITMO, *Software Engineer* Dec 2021 – Jun 2023
Created and developed the open-source framework [GEFEST](#).

PUBLICATIONS

- **N. Starodubcev**, I. Drobyshevskiy, I. Sudakov, D. Baranchuk
[Registers Matter for Pixel-Space Diffusion Transformers](#). *arXiv preprint, 2026*
- **N. Starodubcev**, D. Pakhomov, Z. Wu, I. Drobyshevskiy, Y. Liu, Z. Wang, Y. Zhou, Z. Lin, D. Baranchuk
[Rethinking Global Text Conditioning in Diffusion Transformers](#). *ICLR 2026*

- **N. Starodubcev**, D. Kuznedelev, A. Babenko, D. Baranchuk
[Scale-wise Distillation of Diffusion Models](#). [ICLR 2026](#)
Note: Contributed a new [scheduler](#) to the Hugging Face diffusers library.
- **N. Starodubcev**, M. Khoroshikh, A. Babenko, D. Baranchuk
[Invertible Consistency Distillation for Text-Guided Image Editing in Around 7 Steps](#).
[NeurIPS 2024](#)
Note: Integrated into [YandexART](#), giving a 4x speedup.
- **N. Starodubcev**, D. Baranchuk, A. Fedorov, A. Babenko
[Your Student is Better Than Expected: Adaptive Teacher-Student Collaboration for Text-Conditional Diffusion Models](#). [CVPR 2024](#)
- **N. Starodubcev**, D. Baranchuk, V. Khrulkov, A. Babenko
[Towards Real-Time Text-Driven Image Manipulation with Unconditional Diffusion Models](#).
arXiv preprint, 2023
- **N. Starodubcev**, N. Nikitin, E. Andronova, K. Gavaza, D. Sidorenko, A. Kaluzhnaya
[Generative Design of Physical Objects Using Modular Framework](#).
[Engineering Applications of Artificial Intelligence, 2023](#)

TEACHING EXPERIENCE

Yandex School of Data Analysis, *Lecturer*

Apr – May 2025

Delivered four lectures for the course “Visual Generative Modeling” on modern visual generative models.

Yandex School of Data Analysis, *Lecturer*

Nov 2024

Taught the “CV Week” course on deep understanding of diffusion models. Lecture and seminar ([1](#), [2](#)) delivered in Russian.

AI 360: Yandex and SBER, *Mentor*

Feb 2024

Mentored students from top-tier Russian universities ([HSE](#), [MIPT](#)) on the personalization of diffusion models.